Lecture 1

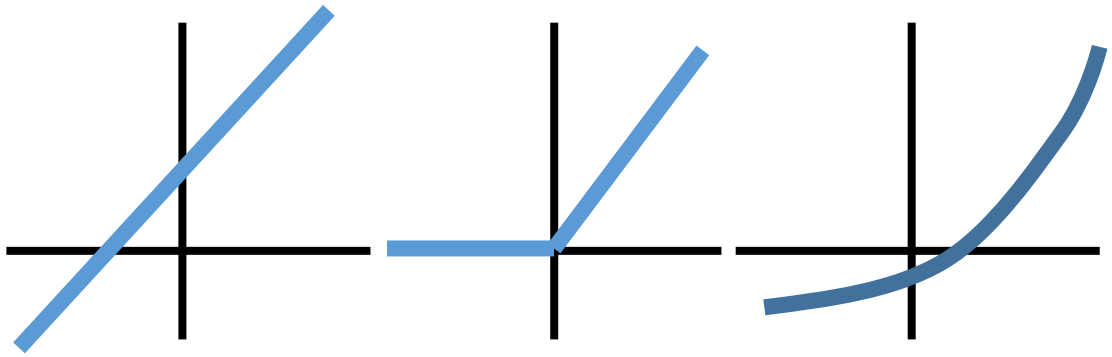
Optimization

- Books:
  - Optimization for Engineering Design: Kalyanmoy Deb
  - Algorithms for Optimization : Mykel J. Kochenderfer and Tim A. Wheeler
    - https://algorithmsbook.com/optimization/files/optimization.pdf
  - Engineering Design Optimization: Joaquim R. R. A. Martins, Andrew Ning
  - https://flowlab.groups.et.byu.net/mdobook.pdf

Single vs multi variable functions:

Functions $f: \mathbb{R} \to \mathbb{R}$ that depend on a single decision variable.

- $f(x) = ax + b, x \in \mathbb{R}$
- $f(x) = ax^2 + bx, x \in \mathbb{R}$
- $f(x) = \max(0, x), x \in \mathbb{R}$

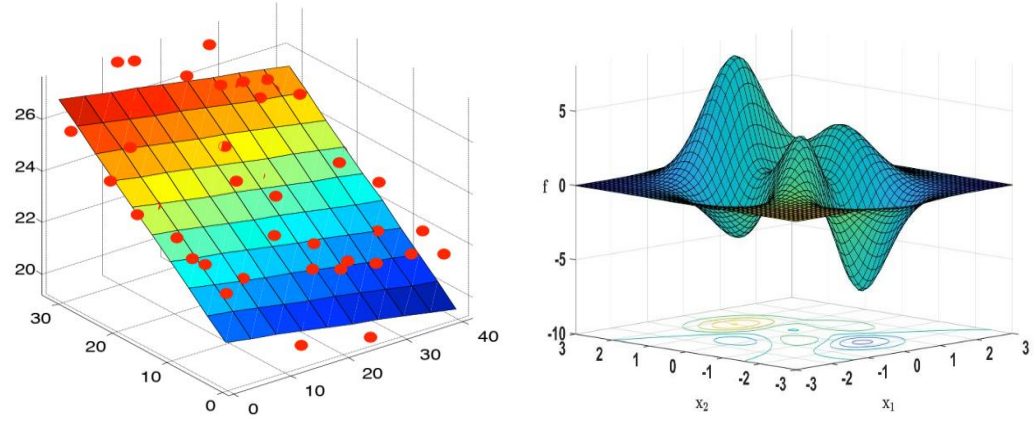


$f: \mathbb{R} \to \mathbb{R}$

- $x \in \mathbb{R}^n \Rightarrow x = (x_1, x_2, x_3, \dots, x_n)$, where $x_i \in \mathbb{R}$
- $x \in \mathbb{R}^3 \Rightarrow x = (x_1, x_2, x_3)$, where $x_i \in \mathbb{R}$
  - $x = (2.4, 5.3, 1, 7)$

Functions $f: \mathbb{R}^n \to \mathbb{R}$ that depend on two or more $(n \geq 2)$ decision variables.

- $f(x_1, x_2, .., x_n) = \sum_{i=1}^{n} a_i x_i + b$
- $f(x_1, x_2) = x_1^2 + x_2^2$



$f: \mathbb{R}^2 \to \mathbb{R}$

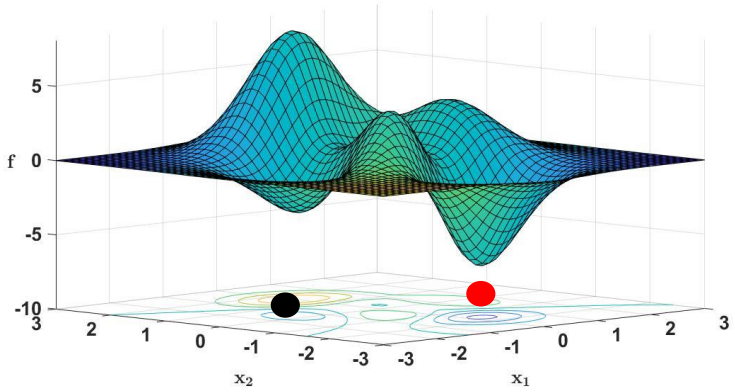
- $x \in \mathbb{Z}^n \Rightarrow x = (x_1, x_2, x_3, \dots, x_n)$, where $x_i \in \mathbb{Z}$
- $x \in \mathbb{Z}^3 \Rightarrow x = (x_1, x_2, x_3)$, where $x_i \in \mathbb{Z}$
  - $x = (0,0,1), x \in \mathbb{Z}^3$, where $x_i \in \{0,1\}$

General frameworks of optimization

Unconstrained optimization:

$$x^* = \arg\min_{x \in \mathbb{R}^n} / \max f(x)$$

where, $f: \mathbb{R}^n \to \mathbb{R}$



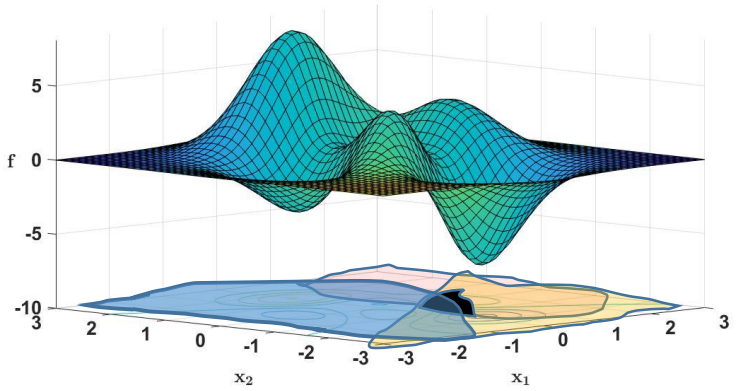
Constrained optimization:

$$x^* = \arg\min_{x \in \mathbb{R}^n} / \max f(x),$$

Subject to

$$g_i(x) \leq 0, i = 1,2, \dots, m$$
$$h_i(x) = 0, i = 1,2, \dots, p$$

where, $f: \mathbb{R}^n \to \mathbb{R}$, $g_i: \mathbb{R}^n \to \mathbb{R}$, $h_i: \mathbb{R}^n \to \mathbb{R}$



Continuous vs discrete optimization

Optimization of functions where variables can take any value within a **continuous** range.

$$x^* = \arg\min_{x \in \mathbb{R}^n} / \max f(x)$$
subject to
$$g_i(x) \leq 0, h_j(x) = 0$$

$$\text{Min } f(x,y) = 2x^2 + 3y^2 + xy$$
Subject to
$$g_1(x,y) = x + y - 1 \leq 0$$
$$g_2(x,y) = -x \leq 0$$
$$x, y \in \mathbb{R}$$

Linear programing (Resource allocation, supply chain)
Non linear optimization
Gradient-Based Training (Loss functions, regression, logistic regression, SVM, classification, Neural Networks)
Hyper parameter Optimization Using Continuous Search Methods:
Continuous Reinforcement Learning (DDPG, PPO etc.)

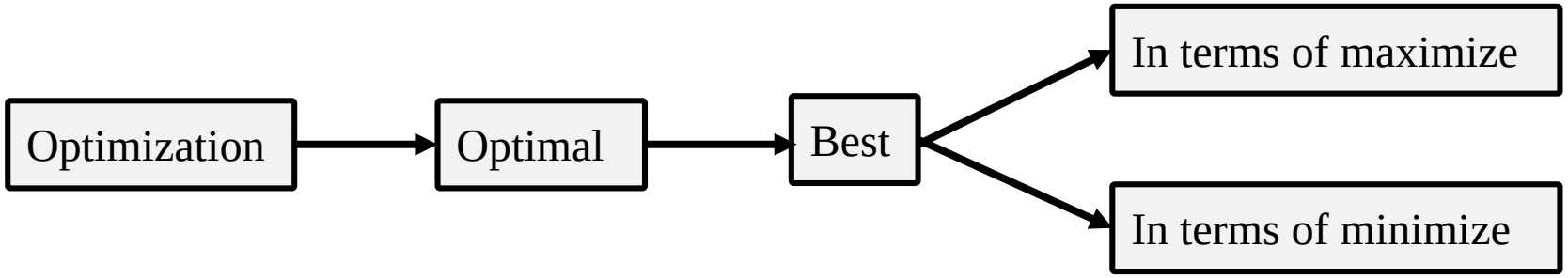
Optimization problems where the decision variables are **discrete**, often integers.

$$x^* = \arg\min_{x \in \mathbb{Z}^n} / \max f(x)$$
subject to
$$g_i(x) \leq 0, h_j(x) = 0$$

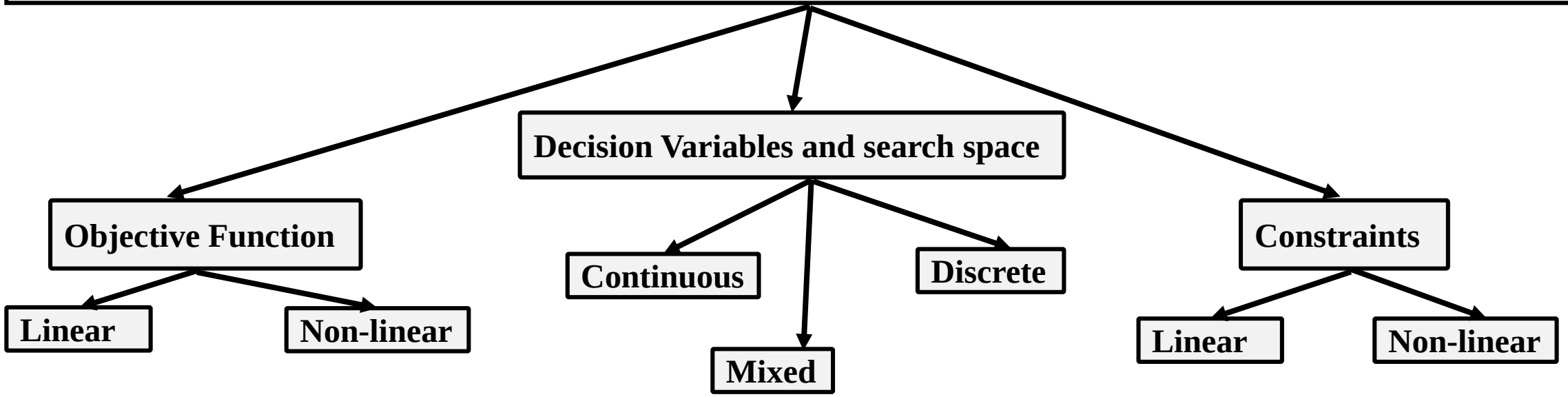
Knapsack problem
$$\text{Max } f(x_1, x_2, x_3) = 60x_1 + 100x_2 + 120x_3$$
Subject to
$$10x_1 + 20x_2 + 30x_3 \leq 50,$$
$$x_i \in \{0,1\}, i = 1,2,3$$

Feature selection
Hyper parameter tuning
Combinatorial optimization (recommendation systems or multi-agent systems.)
Discrete Neural Architecture
TSP for route planning

Optimization:



Optimization is the process of finding the **best** solution from a set of **feasible solutions**, aiming to either **minimize** or **maximize** an **objective function** within a defined **search space**.



Norms:

- Let two points be $x, y \in \mathbb{R}^n$. The distance between them is defined using a norm as:

$$d(x, y) = \| x - y \|$$

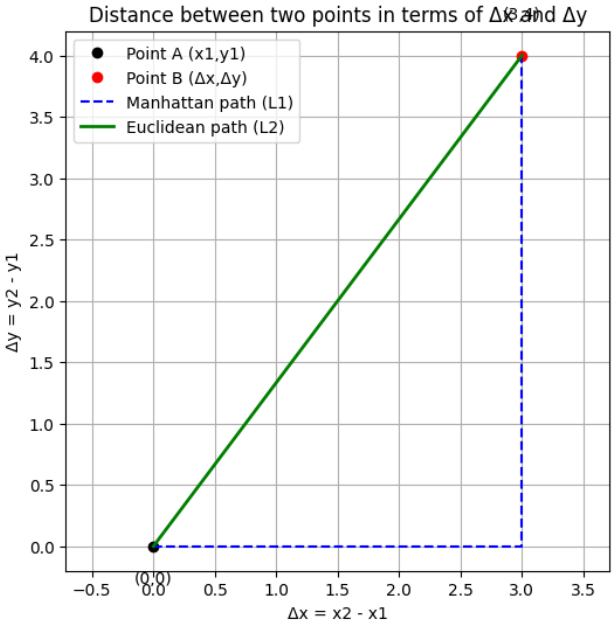
  - l_1 distance (Manhattan distance):

$$\| x - y \|_1 = \sum_{i=1}^{n} | x_i - y_i |$$

  - l_2 distance (Euclidean distance):

$$\| x - y \|_2 = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

    - Let $x = (1,2)$ and $y = (4,6)$ then
      - $\| x - y \|_1 = |4 - 1| + |6 - 2| = 7$
      - $\| x - y \|_2 = \sqrt{(1 - 4)^2 + (2 - 6)^2} = 5$



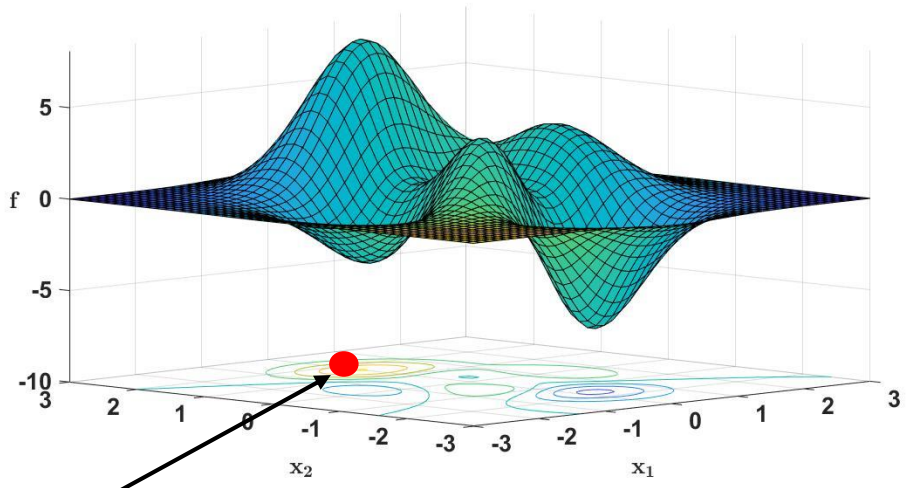
- Norm: Tool to measure the size (magnitude) of vectors.
- l_1 norm (Manhattan Norm or Taxicab Norm)
  - $\| a \|_1 = \sum_{i=1}^{n} | a_i | = |a_1| + |a_2| + \cdots .. + |a_n|, a \in \mathbb{R}^n$
  - If $a = (3, -4, -6)^T$ then $\| a \|_1 = \sum_{i=1}^{3} | a_i | = |3| + |-4| + |-6| = 13$
    - Encourages sparsity (used in Lasso regression, Feature selection).
    - Less sensitive to large values (outliers).

- l_2 *norm* (Euclidean Norm)
  - $\|a\|_2 = \sqrt{\sum_{i=1}^{n} a_i^2} = \sqrt{(a_1)^2 + (a_2)^2 + \cdots + (a_n)^2} = \sqrt{a^T a}, a \in \mathbb{R}^n$
  - If $a = (3, -4, -6)^T$ then $\| a \|_2 = \sqrt{(a_1)^2 + (a_2)^2 + (a_3)^2} = \sqrt{9 + 16 + 36}$
    - Encourages smooth solutions (used in Ridge regression).
    - Penalizes large values more than l_1.

General framework of unconstrained optimization

Let $f: \mathbb{R}^2 \to \mathbb{R}$,

$$\arg\min_{x \in \mathbb{R}^2} f(x_1, x_2) = 3(1-x_1)^2 e^{(-(x_1^2)-(x_2+1)^2)} - 10\left(\frac{x_1}{5} - x_1^3 - x_2^5\right) e^{-(x_1^2+x_2^2)} - \frac{1}{3} e^{(-(x_1+1)^2-x_2^2)}$$

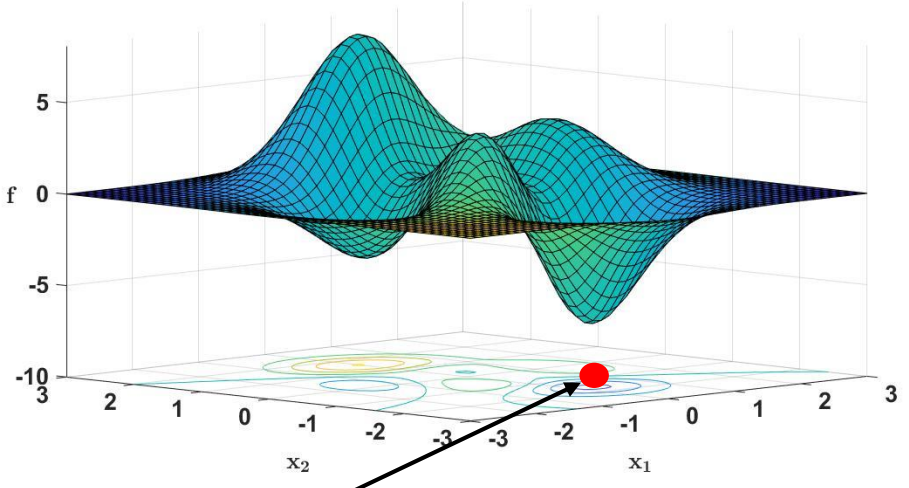


$x^* \in \Omega$ is a local minimizer if

$f(x^*) \leq f(x), \forall x \in B_\delta(x^*)$

Where,

$B_\delta(x^*) = \{x \mid \| x - x^* \|_2 \leq \delta\}$



$x^* \in \Omega$ is a global minimizer if

$f(x^*) \leq f(x), \forall x \in \Omega$

Gradient

- **Derivative:** measuring rate at which a function changes with respect to its independent variable
  - For one dimensional function f: $\mathbb{R} \rightarrow \mathbb{R}$,

$$\frac{d}{dx} f(x) = \lim_{\alpha \to 0} \frac{f(x+\alpha) - f(x)}{\alpha}$$

- For higher dimensional function f: $\mathbb{R}^n \rightarrow \mathbb{R}$,

$$\frac{\partial}{\partial x_i} f(x_1, x_2, \dots, x_n) = \lim_{\alpha \to 0} \frac{f(x_1, x_2, \dots, x_i + \alpha, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\alpha}$$

  - Alternate definition:

$\frac{\partial}{\partial x_i} f(x) = \lim_{\alpha \to 0} \frac{f(x+\alpha e_i) - f(x)}{\alpha}$, where e_i is the ith standard basis vector

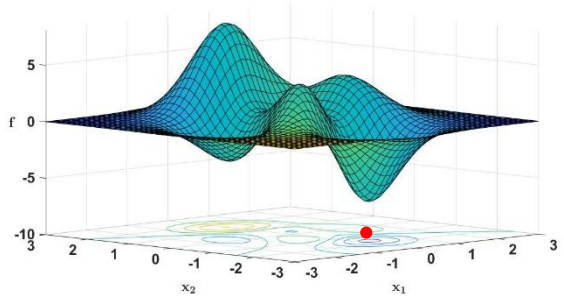
- Gradient:
  - Multivariable generalization of derivative
  - A vector whose components are partial derivatives in each directions

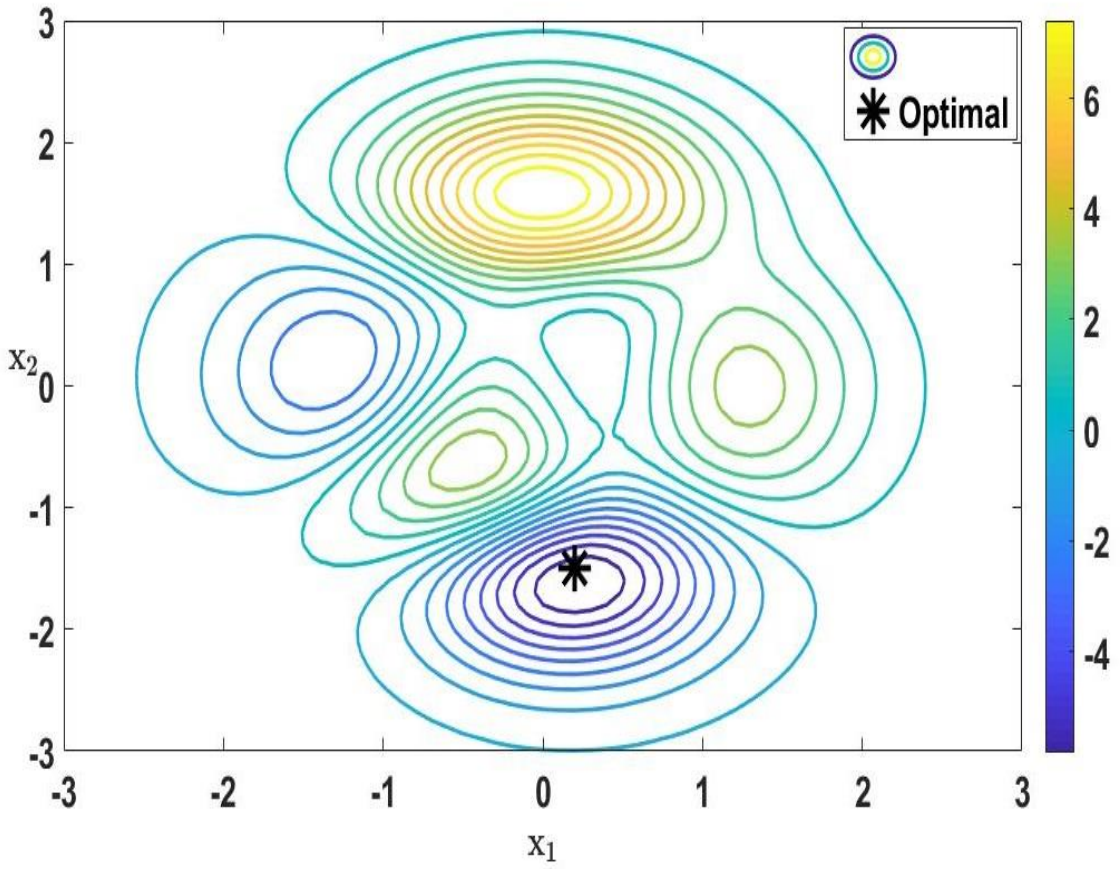
$$\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n}\right]^T$$

Gradient: examples

- $f(x,y) = 3x^2 + 2xy + y^2$
  - $\frac{\partial f}{\partial x} = 6x + 2y$
  - $\frac{\partial f}{\partial y} = 2x + 2y$
- The gradient vector $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]^T = [6x + 2y, 2x + 2y]^T$
- At point (2,3):
  - $\nabla f = [18,10]^T$
  - The vector $\nabla f = [18,10]^T$ at the position (2,3) indicates in the direction where the function f increases most rapidly.
- At point (3,1):
  - $\nabla f = [20,8]^T$
  - The vector $\nabla f = [20,8]^T$ at the position (3,1) indicates in the direction where the function f increases most rapidly.
  - **Moving in gradient vector direction, fastest increase in the function's value will occur.**

Contour





Dot product

- **Dot product (geometrical interpretation):**
- **(Magnitude and Angle Based)**
- Two vectors A, B $\in \mathbb{R}^n$.
- Let $A = [a_1, a_2, \dots., a_n]^T$ and $B = [b_1, b_2, \dots., b_n]^T$. Then

$$A.B = \| A \| \| B \| cos(\theta)$$

- Here $\| A \| = \sqrt{(a_1^2 + \cdots + a_n^2)}$, $\| B \| = \sqrt{(b_1^2 + \cdots + b_n^2)}$ are length of the vectors and θ is angle between them

- **Dot product (Algebraic interpretation):**
- Component-wise Formula
- Two vectors A, B $\in \mathbb{R}^n$.
- Let $A = [a_1, a_2, \dots., a_n]^T$ and $B = [b_1, b_2, \dots., b_n]^T$. Then

$$A.B = \boldsymbol{A^T B} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

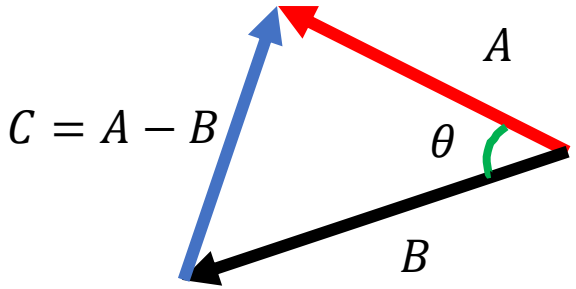
- **Equivalence in both definition:**
- **For same vector**
  - $A = [a_1, a_2, \dots., a_n]$
  - $A.A = \| A \| \| A \| cos(0) = \| A \|^2 = (a_1^2 + a_2^2 + \cdots a_n^2) = A^T A$
- **For different vectors**
  - **Law of cosine:** $\| C \|^2 = \| A \|^2 + \| B \|^2 - 2 \| A \| \| B \| cos(\theta)$ …….(1)
  - Now, $\| C \|^2 = C.C = (A - B)(A - B)$

$$\| C \|^2 = A.A - A.B - B.A + B.B$$

$$\| C \|^2 = \| A \|^2 + \| B \|^2 - 2A.B \quad \text{…….(2)}$$

Comparing (1) and (2),

$$A.B = \| A \| \| B \| cos(\theta)$$



Cosine Similarity:

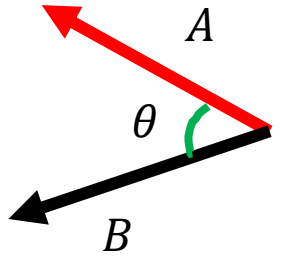
- **Cosine Similarity :**
- **(Magnitude and Angle Based)**
- The dot product depends on both vector magnitude and direction.
- To compare **only the directions** of two vectors, we normalize the dot product.
- **Cosine Similarity** $= \frac{\boldsymbol{A^T B}}{||\boldsymbol{A}||||\boldsymbol{B}||} = \frac{||\boldsymbol{A}||||\boldsymbol{B}||\boldsymbol{cos\theta}}{||\boldsymbol{A}||||\boldsymbol{B}||} = \boldsymbol{cos\theta}$
- **Dot product range:** $(-\infty, \infty)$
- **Cosine Similarity range:** $[-1,1]$

- Projection of vectors
- Orthogonality testing
- Gradient and contour analysis
- Least squares optimization
- Linear kernel in SVM
- Cosine similarity is one of the most important similarity measures used in NLP.
- Measure Semantic Similarity in Large Language Models (LLMs).
- Similarity search and document embeddings

Dot product contd.

Measure of similarity:

- If $\theta = 0^o \Rightarrow A.B$ positive and maximal. A and B have same direction, indicating maximum directional alignment.
- If $\theta = 90^o \Rightarrow A.B = 0$. A and B are perpendicular, indicating no directional correlation.
- If $\theta = 180^o \Rightarrow A.B$ negative and maximal. A and B are opposite direction , indicating maximum directional dissimilarity (anti-alignment).



Measure of similarity:

- $A = [2,3,4]^T, B = [4,1,2]^T$. Then $A^T.B$= (2·4) + (3·1) + (4·2) = 8 + 3 + 8 = 19
- $A = [2,3,4]^T, B = [2,4,2]^T$. Then $A^T.B = (2 \cdot 2) + (3 \cdot 4) + (4 \cdot 2) = 4 + 12 + 8 = 24$
- $A = [1,2,-3]^T, B = [-2,1,2]^T$. Then $A^T.B = (1 \cdot -2) + (2 \cdot 1) + (-3 \cdot 2) = -2 + 2 - 6 = -6$
- $A = [1,2,0]^T, B = [-2,1,1]^T$. Then $A^T.B = (1 \cdot -2) + (2 \cdot 1) + (0 \cdot 1) = -2 + 2 + 0 = 0$

- ***Dot Product: Measures both magnitude and directional alignment.***
- ***Cosine Similarity: Measures directional similarity only (independent of vector magnitude).***
  - $\cos\boldsymbol{\theta = 1}$***: Same direction (maximum similarity)***
  - $\cos\boldsymbol{\theta = 0}$: ***Orthogonal (no directional correlation)***
  - $\cos\boldsymbol{\theta = -1}$: ***Opposite direction (maximum dissimilarity)***

Other Similarity Measure:

- **Manhattan Similarity**
- **Pearson Correlation**
- **Jaccard Similarity**
- **Mahalanobis Similarity**

Unit vector and norms:

- **Unit vector:**
- A unit vector is a vector with a magnitude of 1.
- Direction Representation: To represent direction without regard to magnitude.
- Normalizing input vectors to unit length.
- $v = (3,4)$
- $\hat{v} = \frac{v}{\|v\|} = \frac{(3,4)}{\sqrt{(3^2+4^2)}} = \frac{(3,4)}{5} = (0.6,0.8)$

- Norm: Tool to measure the size (magnitude) of vectors.
- l_1 norm (Manhattan Norm or Taxicab Norm)
  - $\| a \|_1 = \sum_{i=1}^{n} |a_i| = |a_1| + |a_2| + \cdots .. + |a_n|, a \in \mathbb{R}^n$
  - If $a = (3, -4, -6)^T$ then $\| a \|_1 = \sum_{i=1}^{3} |a_i| = |3| + |-4| + |-6| = 13$
    - Encourages sparsity (used in Lasso regression, Feature selection).
    - Less sensitive to large values (outliers).

- l_2 *norm* (Euclidean Norm)
  - $\|a\|_2 = \sqrt{\sum_{i=1}^{n} a_i^2} = \sqrt{(a_1)^2 + (a_2)^2 + \cdots + (a_n)^2} = \sqrt{a^T a}, a \in \mathbb{R}^n$
  - If $a = (3, -4, -6)^T$ then $\| a \|_2 = \sqrt{(a_1)^2 + (a_2)^2 + (a_3)^2} = \sqrt{9 + 16 + 36}$
    - Encourages smooth solutions (used in Ridge regression).
    - Penalizes large values more than l_1.

Directional derivative

- Partial derivative (recall):

$$\frac{\partial}{\partial x_i} f(x) = \lim_{\alpha \to 0} \frac{f(x+\alpha e_i) - f(x)}{\alpha}$$, where e_i is the ith **standard basis vector**

  - $\frac{\partial}{\partial x_i} f(x)$ quantifies the rate of change of f in **direction e_i**.

- Directional derivative:

$$D_d f(x) = \lim_{\alpha \to 0} \frac{f(x+\alpha d) - f(x)}{\alpha}$$, where d is **any unit vector**

  - $D_d f(x)$ quantifies the rate of change of the f in d^{th}direction.

$$D_d f(x) = \nabla f(x^*)^T d$$

$$\nabla f(x^*)^T d = \text{Lim}_{\alpha \to 0} \left\{ \frac{f(\boldsymbol{x}^* + \alpha d) - f(\boldsymbol{x}^*)}{\alpha} \right\}$$

$$f(x^* + \alpha d) = f(x^*) + \alpha \nabla f(x^*)^T d$$

Unit vector only focuses on the rate of change in the specified direction, independent of the vector's length.

Directional derivative

- Given the function $f(x,y) = x^2 + y^2$, calculate the directional derivative of f at the point (1,2) in the direction of the vector $v = (3,4)$.
- Gradient: $\nabla f(x,y) = (2x, 2y)$
- $\nabla f(x,y)$ at $(1,2)$:
  - $\nabla f(1,2)$=(2,4)
- Direction vector $d = (3,4)$
- $\hat{d} = \frac{d}{\|d\|} = \frac{(3,4)}{\sqrt{(3^2+4^2)}} = (\frac{3}{5}, \frac{4}{5})$

Now, $D_d f(x) = \nabla f(x^*)^T \hat{d} = (2,4)^T \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{6}{5} + \frac{16}{5} = \frac{22}{5} = 4.4$

- At the point (1,2), in the direction of d=(3,4) the function $f(x,y)$ (the surface) is increasing at a rate of 4.4 units per unit distance in that direction.

Why the gradient is the direction of greatest ascent?

- Let d^* is the largest directional derivative at x^*
- We seek such optimal d such that

$$d^* = \arg\max_{d:\, \|d\|_2=1} \nabla f(\boldsymbol{x}^*)^T d$$

$$d^* = \arg\max_{d:\, \|d\|_2=1} \| \nabla f(x^*) \|_2 \| d \|_2 \cos(\emptyset)$$

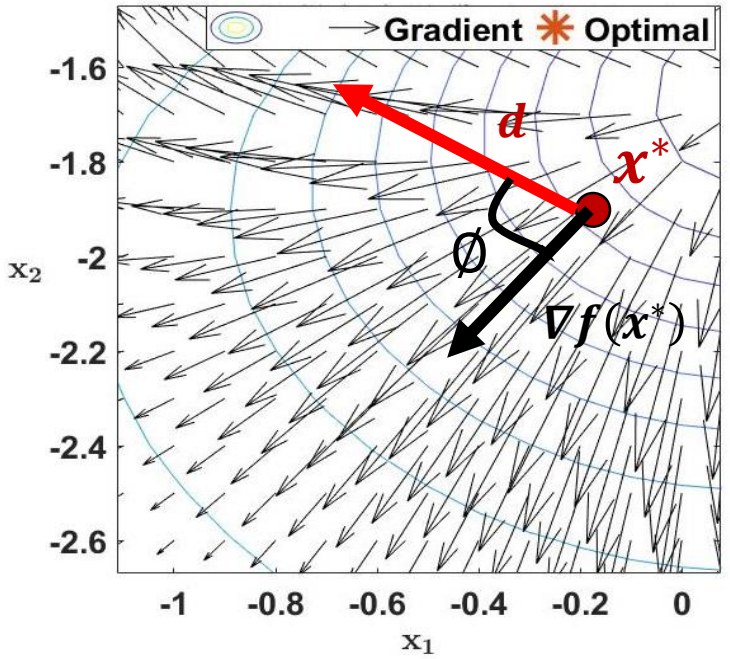
since $\nabla f(\boldsymbol{x}^*)^T d = \| \nabla f(x^*) \|_2 \| d \|_2 \cos(\emptyset)$

$$d^* = \| \nabla f(x^*) \|_2 \arg\max_{d:\, \|d\|_2=1} \| d \|_2 \cos(\emptyset)$$

$$d^* = \| \nabla f(x^*) \|_2 \arg\max_{d:\, \|d\|_2=1} \cos(\emptyset)$$

$$d^* = \| \nabla f(x^*) \|_2$$

$\nabla f(x^*)$ is the direction of the greatest ascent.

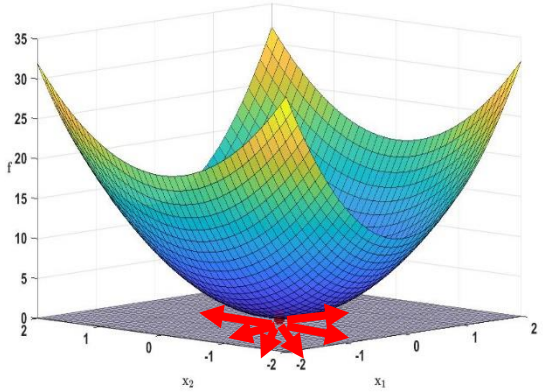


Decent directions: directional derivative as a tool

- If $\boldsymbol{x}^*$ is the optimal solution (minimizer) then

$$\nabla f(\boldsymbol{x}^*)^T d = Lim_{\alpha \to 0} \underbrace{\left\{ \frac{f(\boldsymbol{x}^* + \alpha d) - f(\boldsymbol{x}^*)}{\alpha} \right\}}_{> 0, \forall d}$$

$$\Rightarrow \nabla f(\boldsymbol{x}^*)^T d > 0, \forall d$$



If $\boldsymbol{x}^*$ is not optimal solution (minimizer) then we need a **direction** d such that $f(\boldsymbol{x}^* + \alpha d) - f(\boldsymbol{x}^*) < \mathbf{0} \Rightarrow f(\boldsymbol{x}^* + \alpha d) < f(\boldsymbol{x}^*)$

$$\nabla f(\boldsymbol{x}^*)^T d = Lim_{\alpha \to 0} \underbrace{\left\{ \frac{f(\boldsymbol{x}^* + \alpha d) - f(\boldsymbol{x}^*)}{\alpha} \right\}}_{< 0, \text{ for some } d}$$

$\Rightarrow \nabla f(\boldsymbol{x}^*)^T d < 0 \Rightarrow d$ is the decent direction

$$\nabla f(\boldsymbol{x}^*)^T d = 0 \Rightarrow ?$$

Hessian

- Hessian organizes all second-order partial derivatives of a scalar-valued function $f: \mathbb{R}^n \to \mathbb{R}$
- **Quadratic approximation** of the function $f: \mathbb{R}^n \to \mathbb{R}$

$$H = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & & \vdots & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

- Each element of the Hessian matrix is a second-order partial derivative, indicating how the function's slope (gradient) changes as you move in different directions.

Hessian: **Curvature** measurement around a point

Curvature: degree to which a curve or surface deviates from being straight or flat

- The Hessian is guaranteed to be symmetric only when the mixed second derivatives are continuous (Clairaut's (Schwarz's) theorem),
  - $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$
- (contains real eigenvalues, **spectrum theorem** guarantees that n real eigenvalues, n orthogonal eigenvectors)
- **Quadratic form:** $(x^T A x)$, where A is a **symmetric** matrix

Quadratic form and Hessian

- **Quadratic form:**
- $Q(x) = (x^T Ax)$, where A is a **symmetric** matrix
  - $x \in \mathbb{R}^n$
  - $A \in \mathbb{R}^{n \times n}$
  - x^T is the transpose of x

- Let $x \in \mathbb{R}^2$, means $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$
- $A \in \mathbb{R}^{2 \times 2}$. For example $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$
- Now $Q(x) = x^T Ax = (x_1, x_2) \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$= (ax_1 + bx_2, bx_1 + cx_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= (ax_1 + bx_2)x_1 + (bx_1 + cx_2)x_2$$

$$= ax_1^2 + 2bx_1x_2 + cx_2^2$$ (Second-degree polynomial (quadratic))

- $x^T Ax$ measures how the quadratic form defined by A behaves in the direction of x.

Quadratic form and Hessian

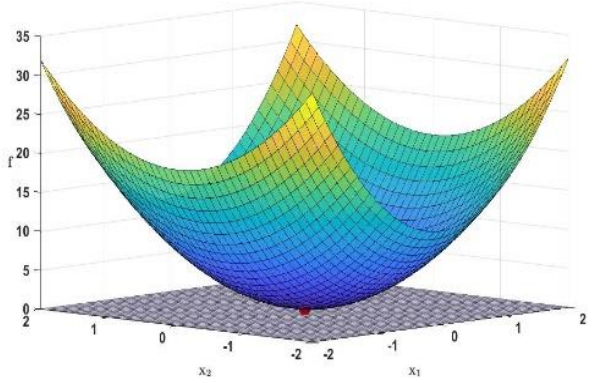
- Let $f(x,y) = 2x^2 + 3xy + y^2$
- $H = \nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$
- $Q(x) = x^T Hx = 4x_1^2 + 6x_1y_1 + 2y_1^2$
- The quadratic form $Q(x) = 4x_1^2 + 6x_1y_1 + 2y_1^2$ describes how the function curves around any point (x_1, y_1).
- The coefficients of x_1^2, x_1y_1 and y_1^2 in the quadratic form reflect the curvature in the x- and y-directions and their interaction.
- Quadratic Forms provide a way to **quadratic approximate** and understand the **local** curvature of a function around a point.
- At $x = (1,1) \Rightarrow Q(x) = 12 > 0 \Rightarrow$ positive curvature of $f(x,y)$ at (1,1) means locally convex (bowl shape)
- At $x = (1,-1) \Rightarrow Q(x) = 0 \Rightarrow f(x,y)$ behavior is neutral in the direction (1,-1), potentially indicating a **saddle point or flat region**.
- If at any $x = \mathrm{x}^* \Rightarrow Q(x^*) < 0 \Rightarrow$ negative curvature of $f(x,y)$ at x^* means locally concave (curves downwards or is at least not bending upwards (locally).)

At any point x:

- $Q(x) = x^T Hx > 0$. $f(x,y)$ have the **positive curvature** at x means **locally convex (bowl shape)**
- $Q(x) = x^T Hx < 0$. $f(x,y)$ have the **negative curvature** at x means **locally concave**
- $Q(x) = x^T Hx = 0$. $f(x,y)$ have **the zero curvature** at x means saddle point or flat region.

Positive definite Hessian

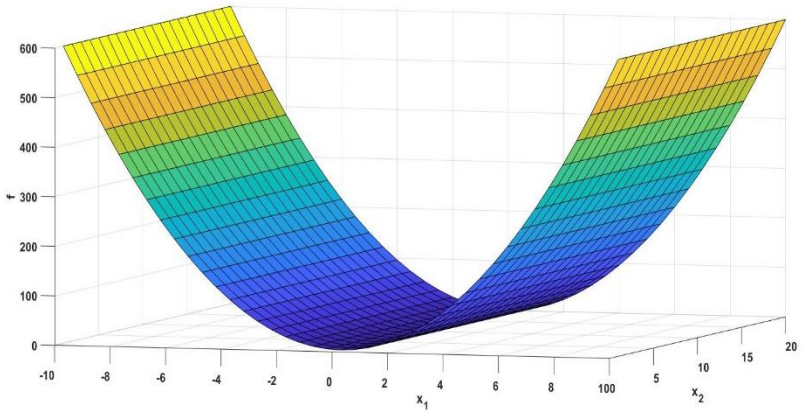
- **Positive definite Hessian**:
- A matrix H is positive definite if all of its eigenvalues λ_i are strictly positive, i.e., $\lambda_i > 0, \forall i$.
- This case, $x^T Hx$ is always positive (positive curvature) for any $x \neq 0$ (i.e., the function curve upwards in all directions)
  - Example: $f(x_1, x_2) = 3x_1^2 + 5x_2^2$
  - $H = \begin{bmatrix} 6 & 0 \\ 0 & 10 \end{bmatrix}$
  - Eigenvalues: $\lambda_1 = 6, \lambda_1 = 10$
  - Eigenvectors: $v_1 = [1,0]^T, v_2 = [0,1]^T$ (orthogonal basis). any direction can be written as a linear combination of v_1 and v_2.
  - $Q(x) = x^T Hx = (x_1, x_2) \begin{bmatrix} 6 & 0 \\ 0 & 10 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 6x_1^2 + 10x_2^2$
  - Since both terms are positive, $Q(x) = x^T Ax > 0$ for any **non-zero** vector x (since both $\lambda_i > 0$ here). The function curves upwards in all directions.



Positive definite (PD) Hessian (H) $\Rightarrow \lambda_i > 0, \forall i \Rightarrow x^T Ax > 0, \forall x \neq 0 \Rightarrow$ Function curves upwards in all directions $\Rightarrow$ strictly convex function

Positive semi definite Hessian

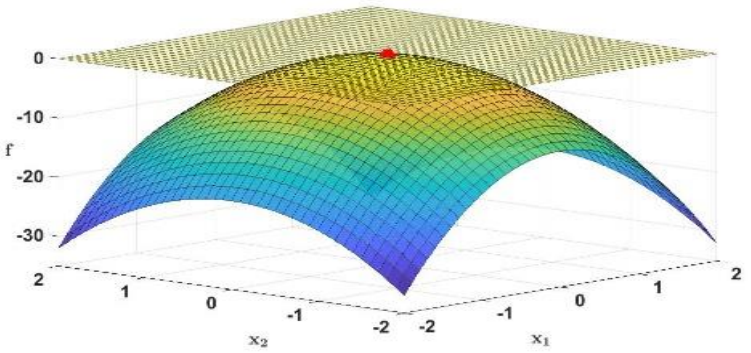
- **Positive semi definite (PSD) Hessian (*H*):**
- A matrix H is positive semi-definite if all of its eigenvalues λ_i are **non-negative**, i.e., $\lambda_i \geq 0$.
- The quadratic form $x^T Hx$ is non-negative, but can be zero for some non-zero vectors x if $\lambda_i = 0$.
  - Example: $f(x_1, x_2) = 6x_1^2$
  - $H = \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix}$
  - $\lambda_1 = 6, \lambda_2 = 0$
  - Eigen vectors: $v_1 = [1,0]^T, v_2 = [0,1]^T$ (orthogonal basis).
  - $Q(x) = x^T Hx = (x_1, x_2) \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 6x_1^2 + 0x_2^2$
  - Here, $Q(x) = x^T Ax \geq 0$. It can be zero when $x_1 = 0$, the function is flat along the x_2-axis (where $\lambda_2 = 0$).
  - The function is convex but flat in the x_2-direction, indicating no curvature along that axis. Along v_1, f increases at a rate of 6, as$\lambda_1 > 0$. Along v_2, f remains flat, as $\lambda_2 = 0$. Other direction: The f increases but at a rate depending on the component along v_1 only.



Positive semi definite Hessian $\Rightarrow \lambda_i \geq 0, \forall i \Rightarrow x^T Ax \geq 0, \forall x \neq 0 \Rightarrow$ Convex function but can have flat regions where the curvature is zero.

Negative definite Hessian:

- **Negative definite Hessian**:
- A matrix H is positive definite if all of its eigenvalues λ_i are strictly negative, i.e., $\lambda_i < 0, \forall i$.
- This case, $x^T Hx$ is always negative (negative curvature) for any $x \neq 0$ (i.e., the function curve downwards in all directions)
  - Example: $f(x_1, x_2) = -x_1^2 - x_2^2$
  - $H = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$
  - $\lambda_1 = -2, \lambda_1 = -2$
  - Eigen vectors: $v_1 = [1,0]^T, v_2 = [0,1]^T$ (orthogonal basis).
  - $Q(x) = x^T Hx = (x_1, x_2) \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -2x_1^2 - 2x_2^2$
  - Since both terms are negative, $Q(x) = x^T Ax < 0$ for any **non-zero** vector x. The function curves downwards in all directions.

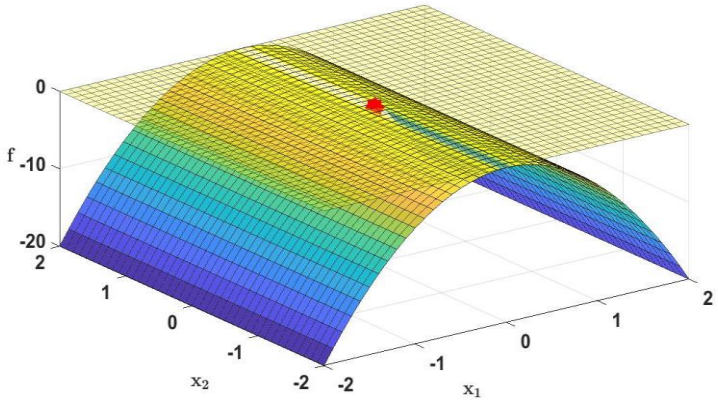


Concave function

Negative definite Hessian (H) $\Rightarrow \lambda_i < 0, \forall i \Rightarrow x^T Ax < 0, \forall x \neq 0 \Rightarrow$
Function curves downwards in all directions ⇒ strictly concave function

Negative semi definite Hessian:

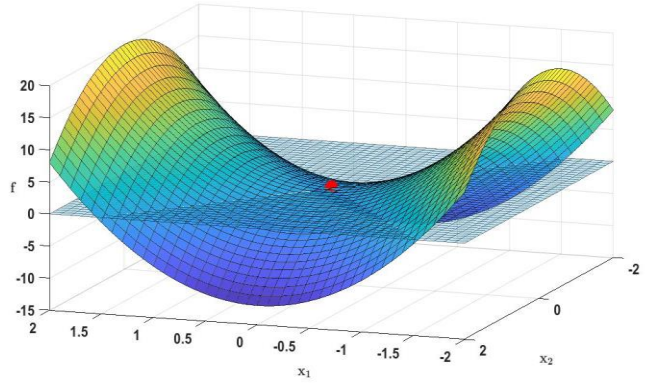
- **Negative definite Hessian**:
- A matrix H is positive definite if all of its eigenvalues λ_i are strictly positive, i.e., $\lambda_i \leq 0, \forall i$.
- This case, $x^T Hx$ is always negative (negative curvature) for any $x \neq 0$ (i.e., the function curve downwards in all directions)
  - Example: $f(x_1, x_2) = -(x_1 + x_2)^2$
  - $H = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}$
  - $\lambda_1 = 0, \lambda_2 = -4$
  - Eigen vectors: $v_1 = [1,1]^T, v_2 = [1,-1]^T$(orthogonal basis)
  - $Q(x) = x^T Hx = (x_1, x_2)\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -2x_1^2 - 4x_1x_2 - 2x_2^2$
  - $Q(x) = x^T Hx = -2x_1^2 - 4x_1x_2 - 2x_2^2 = -2(x_1 + x_2)^2$
  - Here, $x^T Ax \leq 0$. It can be zero when $(x_1 + x_2) = 0$, the function is flat along $(x_1 + x_2) = 0$.



Negative semi definite Hessian $\Rightarrow \lambda_i \leq 0, \forall i \Rightarrow x^T Ax \leq 0, \forall x \neq 0 \Rightarrow$ Function curves downwards in every direction (concave) but may have flat in some directions.

Indefinite Hessian

- **Indefinite Hessian**:
- If the Hessian matrix H has **at least one positive eigenvalue** ($\exists \lambda_i > 0$) and **at least one negative eigenvalue** ($\exists \lambda_j < 0$) , it is called indefinite.
- This case, $x^T H x$ can be positive for some vectors x and negative for others. It means the function curves upwards in some directions and downwards in others.
- Example: $f(x_1, x_2) = x_1^2 - x_2^2$
  - $H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$
  - $\lambda_1 = 2, \lambda_2 = -2$
  - Eigen vectors: $v_1 = [1,0]^T, v_2 = [0,1]^T$ (orthogonal basis).
  - $Q(x) = x^T A x = (x_1, x_2) \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2x_1^2 - 2x_2^2$
  - It indicates that the function has a **saddle point**,
    - with positive curvature in the x_1 direction (i.e., $x_2 = 0$), $f(x_1, 0) = 2x_1^2$
    - with negative curvature in the x_2 direction (i.e., $x_1 = 0$), $f(0, x_2) = -2x_2^2$.



Indefinite Hessian⇒ $\exists \lambda_i > 0$ and $\exists \lambda_j < 0 \Rightarrow x^T A x$ can be positive for some vectors x and negative for others.

Saddle point:

- where the function is neither a local maximum nor a local minimum but has a shape resembling a saddle.
- At a saddle point, the function curves upwards in some directions and downwards in others**.**

All cases

Quadratic form	Condition on $x^T Hx$	Eigenvalues of H	Curvature type	Level curves	Shape
Positive definite	$x^T Hx > 0, \forall x \neq 0$	$\lambda_i > 0, \forall i$	Convex	Ellipses	• Curves upwards in all directions, strict global minimum exists. • Unique global minimum
Positive semi definite	$x^T Hx \geq 0, \forall x$	$\lambda_i \geq 0, \forall i$, with some $\lambda_i = 0$	Convex but can have flat regions where the curvature is zero	Ellipses or degenerate shapes (lines or points)	• Flat regions due to zero eigenvalues • Non unique global minimum
Negative definite	$x^T Hx < 0, \forall x \neq 0$	$\lambda_i < 0, \forall i$	Concave	Ellipses	• Curves downwards in all directions, strict global maximum exists. • Unique global maximum
Negative semi definite	$x^T Hx \leq 0, \forall x$	$\lambda_i \leq 0, \forall i$, with some $\lambda_i = 0$	Concave but can have flat regions where the curvature is zero	Ellipses or degenerate shapes (lines or points)	• Flat regions due to zero eigenvalues • Non unique global maximum
Indefinite	$x^T Hx$ can be positive and negative	Both positive and negative λ_i	Saddle points (mixed curvature)	Hyperbolas or saddle-shaped	Curves upwards in some directions and downwards in others.

Example:

- $f(x,y) = x^3 - 3xy^2$
  - $\frac{\partial f}{\partial x} = 3x^2 - 3y^2$
  - $\frac{\partial f}{\partial y} = -6xy$
  - $\frac{\partial^2 f}{\partial x^2} = 6x$
  - $\frac{\partial^2 f}{\partial y^2} = -6y$
  - $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -6y$
  - $H = \nabla^2 f = \begin{bmatrix} 6x & -6y \\ -6y & -6x \end{bmatrix}$

At point $(x,y) = (1,1) \Rightarrow \begin{bmatrix} 6 & -6 \\ -6 & -6 \end{bmatrix} \Rightarrow \lambda = \pm 6\sqrt{2}$

- At point $(x,y) = (0,1) \Rightarrow \begin{bmatrix} 0 & -6 \\ -6 & 0 \end{bmatrix} \Rightarrow \lambda = \pm 6$
- At point $(x,y) = (0,0) \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \lambda = 0,0$

} Function behaviour at (x,y) ?

Function approximation (Taylor series)

For $f: \mathbb{R} \to \mathbb{R}$, the function approximation around a point x^* is

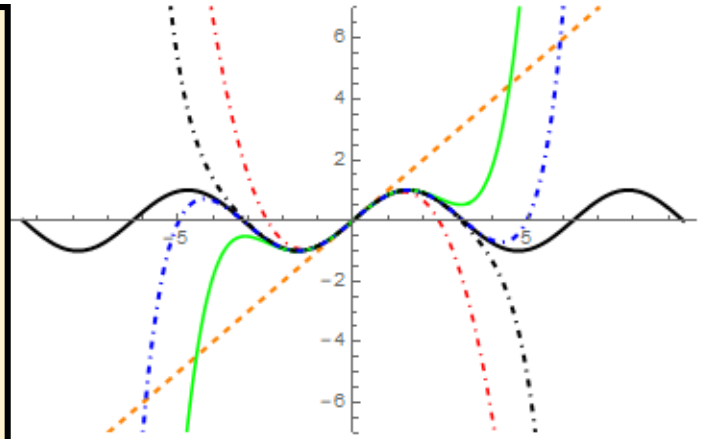
$$f(x) \approx f(x^*) + (x - x^*)f'(x^*) + \frac{(x - x^*)^2}{2!}f''(x^*) + \frac{(x - x^*)^3}{3!}f'''(x^*) + \cdots$$

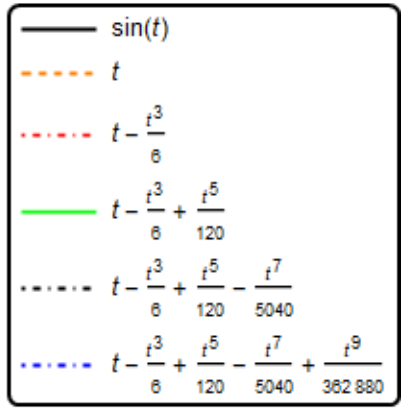
or

$$f(x^* + \alpha) = f(x^*) + \alpha f'(x^*) + \frac{\alpha^2}{2!}f''(x^*) + \frac{\alpha^3}{3!}f'''(x^*) + \cdots$$

- If $f(x) = \sin x$, and $x^* = 0$
- $\sin x \approx \sin 0 + (x - 0)\cos 0 + \frac{(x-0)^2}{2}(-\sin 0) + \frac{(x-0)^3}{6}(-\cos 0) + \cdots$
- $\sin x \approx x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots.$

  or
- $\sin t \approx t - \frac{t^3}{6} + \frac{t^5}{120} - \cdots.$





Taylor series: Gradient, hessian and Function approximation

For $f: \mathbb{R} \to \mathbb{R}$, the function approximation around $x^*$is

$$f(x) = f(x^*) + (x - x^*)f'(x^*) + \frac{(x - x^*)^2}{2!}f''(x^*) + \frac{(x - x^*)^3}{3!}f'''(x^*) + \cdots$$

or

$$f(x^* + \alpha) = f(x^*) + \alpha f'(x^*) + \frac{\alpha^2}{2!}f''(x^*) + \frac{\alpha^3}{3!}f'''(x^*) + \cdots$$

For $f: \mathbb{R}^n \to \mathbb{R}$, the function approximation around $x^*$is

$$f(x) = f(x^*) + (x - x^*)\boldsymbol{\nabla f}(\boldsymbol{x}^*) + \frac{1}{2}(x - x^*)^T\ \boldsymbol{\nabla^2 f}(\boldsymbol{x}^*)(x - x^*) + \cdots$$

or (substitute $x = x^* + \alpha d$)

$$f(x^* + \alpha d) = f(x^*) + \boldsymbol{\alpha \nabla f}(\boldsymbol{x}^*)\boldsymbol{d} + \frac{\alpha^2}{2}\underbrace{d^T\ \boldsymbol{\nabla^2 f}(\boldsymbol{x}^*)d}_{\textbf{Quadratic form } \boldsymbol{x^T H x}} + \cdots$$

Function approximation in a specific point x^*

$$x^* + \alpha d$$

First order approximation:

$$f(x^* + \alpha d) \approx f(x^*) + \alpha \nabla f(x^*)^T d$$

Second order approximation:

$$f(x^* + \alpha d) \approx f(x^*) + \alpha \nabla f(x^*)^T d + \frac{\alpha^2}{2} d^T \nabla^2 f(x^*) d$$

- $\nabla^2 f(x^*)$: local curvature at point x^*
- $d^T \nabla^2 f(x^*) d$: **quadratic form** introduces information about the **curvature** of the function f at the point x^* in d^{th} direction and accounts for the nonlinearity that the linear term doesn't capture
- Real eigenvalues of $\nabla^2 f(x^*)$, condition number
- Second order algorithms: Newton, Quasi Newton (BFGS, trust regions)

Optimality conditions (*Unconstrained case*):

How gradient and hessian works?

Why optimality conditions are needed?

- Identify candidate points for being **local** optima (minima or maxima)**.**
- Optimality conditions are criteria used to determine if a solution is optimal.

First order necessary condition (**FONC**) for local optimality (**Stationarity**):

- At a local optima (minimum or maximum), the gradient of the objective function must be zero. It means, at local optima $x^*, \nabla f(x^*) = 0$

$\nabla f(x^*) = 0$ identify the critical (or stationary) points

- Why is $\nabla f(x^*) = 0$ necessary?
- Suppose x^* is the minimizer.
- Pick any direction d, and any step size α. Then
  $$f(x^* + \alpha d) = f(x^*) + \alpha \nabla f(x^*)^T d + O(\alpha^2)$$
- Rearranging the terms yields
  $$Lim_{\alpha \to 0} \left\{ \frac{f(x^* + \alpha d) - f(x^*)}{\alpha} \right\} = \nabla f(x^*)^T d$$
- Since x^* is the minimizer therefore,
  $$Lim_{\alpha \to 0} \left\{ \frac{f(x^* + \alpha d) - f(x^*)}{\alpha} \right\} \geq 0 \Rightarrow \nabla f(x^*)^T d \geq 0, \forall d$$
- $\nabla f(x^*)^T d \geq 0, \forall d \Rightarrow \nabla f(x^*) = 0$

Second order Optimality:

Second order necessary condition (SONC) for local optimality

- At local minima x^*, it is necessarily $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \geq 0$ (i.e., $\nabla^2 f(x^*)$ is Positive semi definite)
- . If x^* is a local minima of $f(x)$ then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \geq 0$

- Let x^* is the local minima.
- Consider some direction d, and step size $\alpha > 0$. Let takes the second order approximation

$$f(x^* + \alpha d) = f(x^*) + \underbrace{\alpha \nabla f(x^*)^T d}_{=0} + \frac{\alpha^2}{2} d^T \nabla^2 f(x^*)^T d + \frac{\alpha^3}{6} O(\| \alpha \|^3)$$

- $Lim_{\alpha \to 0} \frac{1}{\alpha^2}\{f(x^* + \alpha d) - f(x^*)\} = \frac{1}{2} d^T \nabla^2 f(x^*)^T d + lim_{\alpha \to 0} \frac{\alpha^3}{6} O(\| \alpha \|^3)$
- Since x^* is the local minima, at x^*, $f(x^* + \alpha d) \geq f(x^*)$
- $\underbrace{d^T \nabla^2 f(x^*)^T d}_{\text{Quadratic form}} \geq 0, \forall d \Rightarrow \nabla^2 f(x^*)$ is the positive semi definite

$\nabla^2 f(x^*)$ identify the nature of f around critical point

Second order sufficient condition (SOSC) for local optimality

If f satisfies $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) > 0$ (Positive Definite) on point x^* of its domain then $x^*$is a **strict local minimum** of f.

- Consider some direction d, and step size $\alpha > 0$. Let takes the second order approximation

$$f(x^* + \alpha d) = f(x^*) + \underbrace{\alpha \nabla f(x^*)^T d}_{=0} + \frac{\alpha^2}{2} \underbrace{d^T \nabla^2 f(x^*)^T d}_{>0} + \frac{\alpha^3}{6} O(\| \alpha \|^3)$$

- $$\{f(x^* + \alpha d) - f(x^*)\} = \underbrace{\frac{\alpha^2}{2} d^T \nabla^2 f(x^*)^T d}_{>0} + lim_{\alpha \to 0} \frac{\alpha^3}{6} O(\| \alpha \|^3)$$

- $f(x^* + \alpha d) > f(x^*), \forall d \Rightarrow x^*$ is a strict local minima

- If x^* satisfies, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is Positive definite $\Rightarrow x^*$is a strict local minimum of f
- If x^* satisfies, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is Negative definite $\Rightarrow$ $x^*$is a strict local maximum of f
- If x^* satisfies, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is Indefinite $\Rightarrow x^*$is a saddle point of f

Convexity

First order Convexity:

A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if its domain is a **convex set** and for all x_0, x_1 in **its domain**, and all $\lambda \in [0,1]$, we have

$$f(\lambda x_0 + (1-\lambda)x_1) \leq \lambda f(x_0) + (1-\lambda) f(x_1)$$

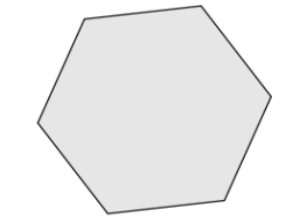
- **Convex set:** A set C is convex if the line segment between any two points in C lies in C, i.e. $\forall x_1, x_2 \in C, \forall \lambda \in [0,1]$

$$\lambda x_1 + (1-\lambda)x_2 \in C$$

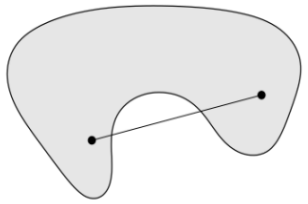
First order Convexity:

$$f(x + \alpha d) \geq f(x) + \nabla f(x)^T d, \ \forall x, d \in \mathbb{R}^n, \alpha > 0$$

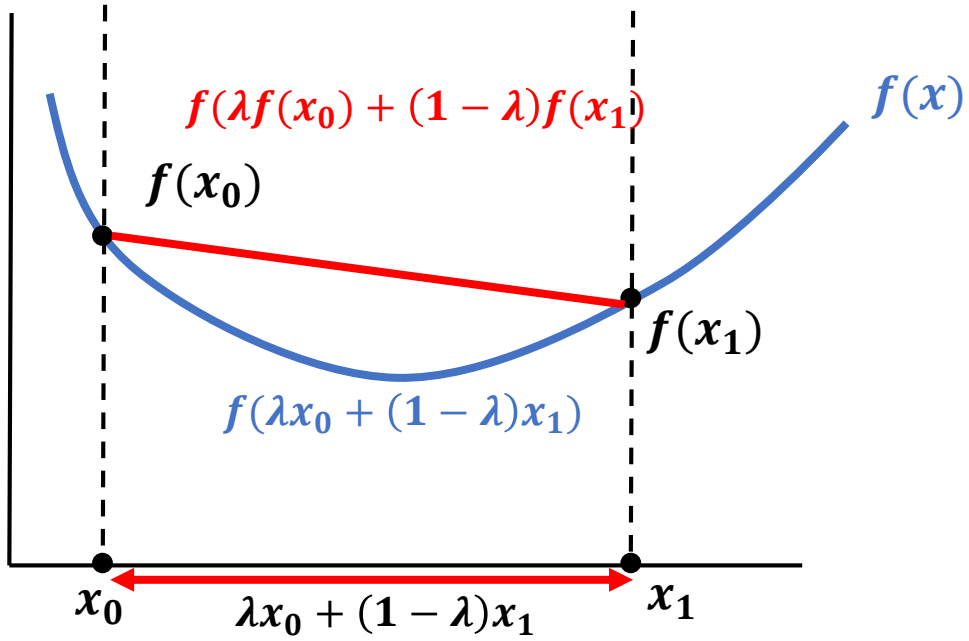
- The function $f(x)$ lies above its linear approximation at any point x in $\mathbb{R}^n$.
- **At a local minima x^*, $\nabla f(x^*) = 0$ it means $f(x^* + \alpha d) \geq f(x^*)\ \forall x, d \in \mathbb{R}^n, \alpha > 0 \Rightarrow x^*$ is the global minima**

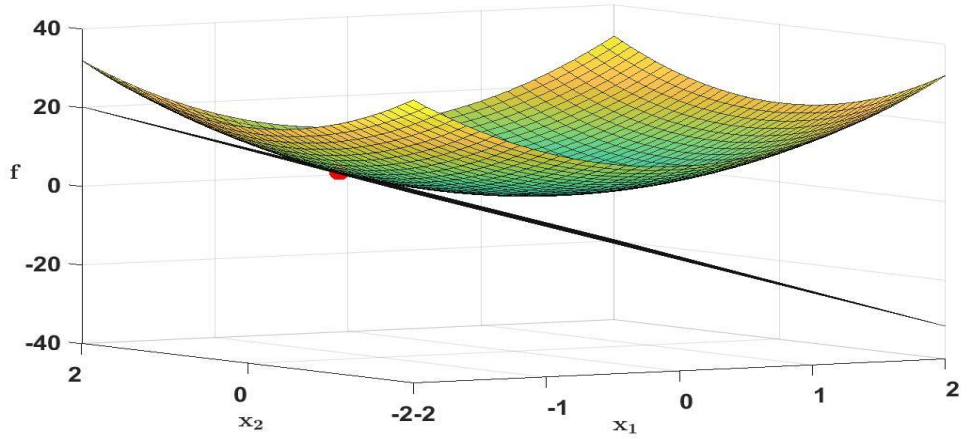


Convex set



Non-convex set





$f(x) = f(x_1, x_2) = 5x_1^2 + 3x_2^2, H = \begin{bmatrix} 10 & 0 \\ 0 & 6 \end{bmatrix}$, *eig=10,6*

Convexity

Second order Convexity:
f is convex over a domain Ω **if and only if** $\nabla^2 f(x) \geq 0, \forall \boldsymbol{x} \in \boldsymbol{\Omega}$

Analytical vs Numerical optimization

Analytical optimization:
Finding the **exact solution** to an optimization problem using mathematical methods (FONC, SONC, SOSC) and derivations.

Merit:
- Exact Solutions
- Efficiency
- No Numerical Errors.

Limitations:
- Complexity
- Closed form requirements
- Local search (convexity helps to make it global)

Numerical optimization:
Numerical optimization uses iterative algorithms to find **approximate solutions** to optimization problems, especially when analytical solutions are not feasible..

Merit:
- Applicability
- Flexibility
- Scalability
- Local and global (genetic algorithm) both kind of algorithms

Limitations:
- Approximate Solutions:
- Sensitivity to Initial Conditions
- Numerical Stability
- Lack of convergence studies